

# **NUMERICAL AND EXPERIMENTAL ANALYSIS OF HEAT TRANSFER IN AN ANNULAR MINICHANNEL USING STAR-CCM+**

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## **ABSTRACT**

The present study combines experiments and CFD simulations to characterize heat transfer during HFE-649 flow in a vertical annular minichannel. The experiments were conducted at a fixed mass flow rate of 8 kg/h for two heater power inputs (Set #1 and Set #2), and the measured inlet fluid temperature, inlet/outlet pressures, and heater power input were used to define the boundary conditions. Based on the experimental conditions, Set #1 corresponds to single-phase flow, whereas Set #2 reaches the subcooled boiling region. A three-dimensional conjugate heat-transfer model was developed in Simcenter STAR-CCM+. For both sets, the simulations were performed using Finite Volume Method (FVM). For Set #2, turbulence effects were represented using a Reynolds-Averaged Navier–Stokes (RANS) turbulence model. The CFD predictions were validated against measured axial wall-temperature profiles at thermocouple locations. Furthermore, local heat transfer coefficients were derived from the CFD fields using the wall heat flux and the wall-to-bulk temperature difference to quantify axial heat-transfer trends. The predicted temperature profiles remained in satisfactory agreement with the measurements for both sets. The resulting heat transfer coefficient distributions exhibit an entrance region with elevated values and downstream development that differs between Set #1 and Set #2, consistent with the higher heat input and the approach to subcooled-boiling conditions in Set #2. The proposed modelling approach supports interpretation of the experimental data and provides a basis for further numerical studies of heat transfer in annular minichannels under controlled heating conditions.

**KEYWORDS:** Annular minichannel, Heat transfer, CFD, Simcenter STAR-CCM+, Dielectric liquid

## **1. INTRODUCTION**

Heat transfer in narrow annular channels and minigaps is of growing interest in the design of compact, high-performance heat exchangers and cooling systems. Applications include thermal management of power electronics, battery cooling plates in electric vehicles, micro-energy systems, and high-precision laboratory equipment. In such systems, the characteristic hydraulic dimensions are of the order of a few millimeters or less, and the flow frequently remains in the low-Reynolds-number regime over a wide range of operating conditions. This leads to a strong coupling between the developing velocity and temperature fields and to a pronounced sensitivity to geometry and boundary conditions.

Over the years, the literature has reported only a limited number of studies addressing heat transfer in annular minichannels, most of them using water as the working fluid. Dirker and Meyer (2002) [1] investigated forced-convection heat transfer in smooth concentric annuli for turbulent water flow, with particular emphasis on the effect of annular diameter ratio on the inner-wall heat transfer coefficient. Using the Wilson-plot method, they proposed a correlation for predicting inner-wall Nusselt numbers; for annular diameter ratios between 1.7 and 3.2 and Reynolds numbers from 4,000 to 30,000, the correlation reportedly matched measured values within about 3%. This work provides a useful single-phase turbulent reference baseline for annular passages, although it does not address dielectric refrigerants or mini-scale gaps.

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Complementary insight into narrow-annulus behavior in the transitional and low-Reynolds-number regimes was provided by Lu and Wang (2008) [2], who studied single-phase water convection in a narrow concentric annulus and demonstrated differences relative to conventional circular tubes. Their experiments covered three flow orientations (horizontal, vertical upward and vertical downward) over a broad Reynolds-number range ( $\sim 10$ – $30,000$ ) in a tube-in-tube test section (hydraulic diameter 4.12 mm; heated length 1500 mm). They reported an earlier laminar–turbulent transition ( $Re \approx 800$ – $1200$ ) and showed that at very low Reynolds numbers ( $Re < 150$ ) axial heat conduction can significantly deteriorate the apparent heat transfer. Importantly for data reduction in small geometries, they also compared two approaches for evaluating heat transfer coefficients and showed that the resulting Nusselt/heat transfer coefficient values can differ substantially, especially for  $Re < 400$ , highlighting the sensitivity of inferred results to methodological choices.

Beyond water-based studies, enhancement concepts in annular mini heat exchangers have also been explored using modified working fluids. Bezaatpour and Goharkhah (2020) [3] investigated an active heat-transfer enhancement strategy for a double-pipe mini heat exchanger (annular passage) by applying an external magnetic field to a magnetic nanofluid, thereby inducing swirl and promoting mixing. Their three-dimensional numerical simulations covered the effects of nanoparticle concentration, Reynolds number and magnetic-field intensity, and indicated that the induced swirl can disrupt the thermal boundary layer and substantially increase convective heat transfer, with reported enhancements up to about 320% under favorable conditions. Although the working fluid differs from dielectric refrigerants, the study illustrates how secondary-flow generation in annular passages can strongly affect heat-transfer trends while introducing comparatively modest pressure-drop penalties.

More recently, Yang et al. (2024) [4] combined experiments and CFD (ANSYS Fluent) to study single-phase water flow in a horizontal mini-annular channel with an annular gap of 3.0 mm. They examined the effects of inlet Reynolds number, imposed heat flux and inlet temperature on thermal–hydraulic performance, and validated their CFD results against measurements and selected literature correlations. In their numerical setup, inlet/heated/outlet sections and the inner tube were included, a uniformly distributed inner-wall heat flux was imposed, and external heat losses were represented via an ambient boundary condition applied at the outer wall. A notable conclusion was that the laminar–turbulent transition may occur earlier than expected in mini-annular configurations, reinforcing the need to carefully assess flow-regime assumptions when interpreting heat-transfer data in small annular gaps.

For dielectric coolants relevant to compact electronics cooling, experimental studies in annular minigaps have been reported for HFE-649 under boiling conditions. Piasecka, Musiał and Piasecki (2019) [5] investigated flow boiling of HFE-649 in an annular minigap formed between an inner metal pipe with an enhanced surface and an outer coaxial glass tube. Heating was provided by a cartridge heater, and thermocouples were used to measure the metal-pipe surface temperature. Local heat transfer coefficients under steady-state conditions were determined using a one-dimensional method for a multilayer cylindrical wall, enabling evaluation of boiling heat-transfer performance for an enhanced-wall annular minigap configuration. In a subsequent study, Hożejowska and Piasecka (2020) [6] examined flow boiling of HFE-649 in a vertically oriented annular minigap with a 2 mm gap, with the specific aim of determining local boiling heat transfer coefficients along the channel. They formulated an axisymmetric, steady inverse heat-conduction problem (with laminar flow assumed in the minigap) and solved it using the Trefftz method, extracting local coefficients via a third-kind boundary condition under saturated boiling. Thermocouple-based heater-surface temperatures and inlet/outlet fluid temperatures supported the reconstruction, and the uncertainty/accuracy of the identified coefficients was quantified.

Taken together, the state-of-the-art indicates that heat transfer in annular minichannels is strongly influenced by inlet conditions, axial development and conjugate heat conduction through surrounding solids. For the low-Reynolds-number flows typical of dielectric liquids, the entrance region may occupy a substantial fraction of the heated length, making local (rather than only average) quantities essential for meaningful model assessment. Reported discrepancies between CFD predictions and experiments are frequently linked to boundary-condition specification, heat losses to the surroundings and uncertainty in near-wall temperature determination in small geometries. Consequently, carefully validated conjugate heat-transfer CFD models

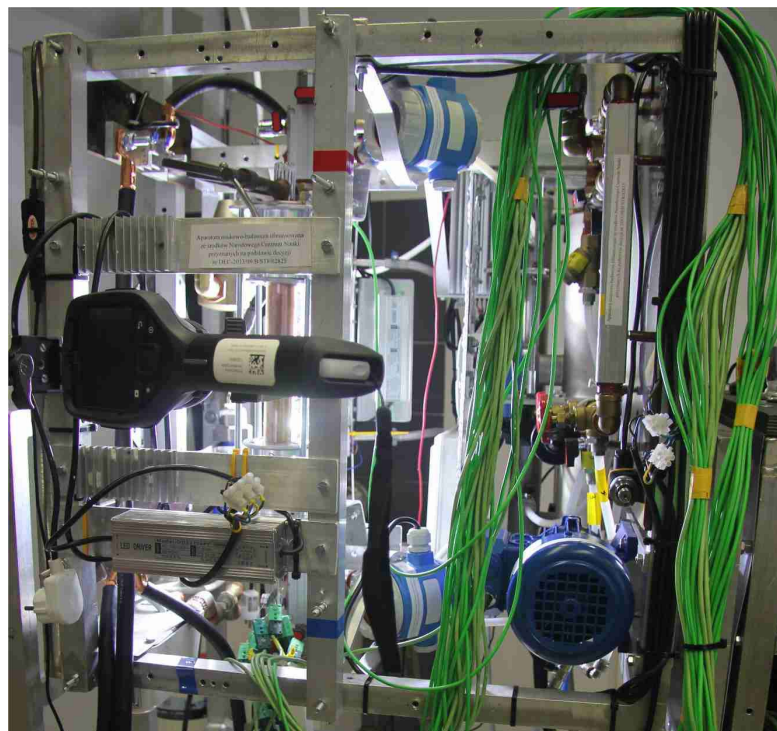
remain important tools for interpreting experiments and supporting design decisions in annular minichannel applications.

On the numerical side, commercial CFD packages such as ANSYS Fluent/CFX, COMSOL Multiphysics and Simcenter STAR-CCM+ (Siemens) are widely used for conjugate heat-transfer analyses in complex geometries. STAR-CCM+ software offers robust meshing and physical modelling capabilities suitable for laminar and turbulent single-phase flows, and can be extended to more advanced multiphase and phase-change simulations when required. When appropriately validated against experiments, STAR-CCM+ has been shown to reproduce detailed temperature fields and heat-transfer trends in compact channel configurations, provided that conjugate effects and heat losses are represented consistently.

The objective of this study is to validate CFD-based conjugate heat-transfer modelling of HFE-649 flow in a vertical annular minichannel against experimental data obtained at a fixed mass flow rate of 8 kg/h for two heater power inputs (Set #1 and Set #2). Set #1 represents single-phase convection, whereas Set #2 corresponds to conditions reaching the subcooled boiling region. The numerical analysis is performed within a conjugate heat-transfer framework using the Finite Volume Method (FVM) for both sets, with a single-phase laminar flow model for Set #1 and a RANS turbulence model for Set #2. The validation focuses on axial wall-temperature distributions measured at thermocouple locations and on local heat-transfer characteristics derived from the CFD fields.

## 2. EXPERIMENTAL SETUP AND TEST PROCEDURE

**2.1. Test loop and working fluid.** The experimental facility is a closed-loop circulation system using HFE-649 as the working fluid. The main components of the loop include the test module with the annular minichannel, a gear pump, a pressure regulator serving as an expansion and compensation vessel, a water-cooled tube-in-tube heat exchanger, a Coriolis mass flow meter, a separator and filters, and pressure transducers and thermocouples located at the inlet and outlet of the test section. The test module was mounted vertically, with the fluid flowing upward through the annular gap. The loop enables independent control of the mass flow rate, system pressure, and heater power input. A view of the experimental setup is shown in Fig. 1. Further details of the facility and the test module are provided in [6,7].



**Fig. 1** A view of the experimental setup.

**2.2. Test module geometry.** The test module (Fig. 2) consists of a concentric glass–copper annulus with an internal cartridge heater:

- outer borosilicate glass tube, inner diameter 31 mm, outer diameter 35 mm;
- inner copper tube, inner diameter 26 mm, outer diameter 28 mm;
- annular minichannel formed between the outer surface of the copper tube and the inner surface of the glass tube, with an annular gap of 1.5 mm and an effective heated length of 200 mm;
- centrally located cartridge heater housed inside the copper tube;
- inlet and outlet heads with appropriate sealing and fluid connectors.



**Fig. 2** A view of the test module.

The cartridge heater consists of a Kanthal resistance wire wound on a ceramic core, enclosed in a stainless-steel sheath. The annular gap between the heater sheath and the copper tube is filled with thermally conductive paste to minimize contact resistance and ensure good thermal coupling. To measure the temperature distribution along the heater, 18 type-K thermocouples (diameter of 0.5 mm) are installed in longitudinal slots machined into the outer surface of the heater sheath. The thermocouples are arranged in a helical pattern, with an axial spacing of 10 mm. Additional thermocouples measure the inlet and outlet fluid temperatures, while pressure transducers monitor the relative pressure at the inlet and outlet of the minichannel.

**2.3. Operating conditions and experimental background.** The experimental campaign yielded two representative steady-state operating points at a constant mass flow rate of 8 kg/h. The heater power input was set to two levels: Set #1 (lower heat-input case) and Set #2 (higher heat-input case). During experiment, atmospheric pressure was 101 kPa while temperature of ambient air 293.15 K.

The system was allowed to reach steady state before recording all measured quantities (electrical heater power input, inlet/outlet fluid temperatures and pressures, and thermocouple readings along the heater surface). From the resulting dataset, representative operating points were selected for detailed CFD analysis, including a case in the single-phase forced-convection regime and a case in the subcooled boiling region.

Flow-regime classification for Set #1 and Set #2 was assessed using the difference between the measured surface temperature (heater-sheath surface temperature measured by thermocouples) and the saturation temperature ( $\Delta T$ ). The local saturation temperature  $T_{sat}$  was evaluated from the local absolute pressure  $p_{abs}$ , obtained by linear interpolation between the measured inlet and outlet pressures along the heated length. Based on  $\Delta T$ , Set #1 corresponds to single-phase flow, whereas Set #2 reaches the subcooled boiling region.

Measured operating conditions of the experimental series, used as boundary conditions in Simcenter STAR-CCM+ software, are listed in Table 1. Wall-temperature measurements from thermocouples located on the heater sheath were used for validation of the results obtained from numerical calculations.

**Table 1.** The characteristics of the experimental series.

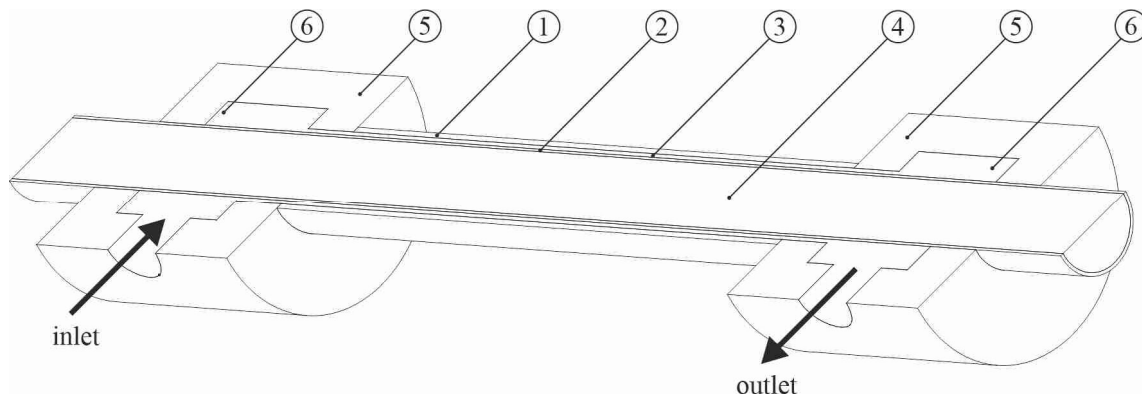
| Main measured operating conditions  | Set #1  | Set #2  |
|-------------------------------------|---------|---------|
| Mass flow rate (kg/s)               | 0.00222 | 0.00222 |
| Fluid temperature at the inlet (K)  | 294.05  | 294.15  |
| Fluid temperature at the outlet (K) | 310.95  | 314.45  |
| Gauge pressure at the inlet (kPa)   | 157     | 161     |
| Gauge pressure at the outlet (kPa)  | 146     | 153     |
| Heater power input (W)              | 72.8    | 91.8    |

It is important to note that the CFD model described in the next section was driven exclusively by measured boundary conditions: the inlet fluid temperature, the inlet and outlet pressures, the mass flow rate, and the heater power input. No measured wall temperatures were imposed in the simulations. Thermocouple measurements on the heater sheath were used exclusively to validate the predicted temperature profiles.

### 3. NUMERICAL MODEL

**3.1. Geometry and computational domain.** The three-dimensional geometry of the test module, including the annular minichannel and surrounding solid components, was created in SOLIDWORKS software and imported into CFD software. Numerical simulations were performed using Simcenter STAR-CCM+ (Version 2020.2.1, Build 15.04.010). The computations were carried out on a PC equipped with an Intel Core i9 CPU (24 cores, 3.50 GHz) and 256 GB RAM. Detailed information on the computational methodology in Simcenter STAR-CCM+ is provided in [8].

The three-dimensional geometry of the test module is illustrated in Fig. 3.



**Fig. 3** The three-dimensional digital model of the test module for numerical calculations in Simcenter STAR-CCM+, created in SOLIDWORKS; legend: 1 – glass tube, 2 – copper tube, 3 – annular minichannel, 4 – cartridge heater, 5 – module head with sealing, 6 – fluid nozzle.

The computational domain comprises:

- the fluid region representing the vertical annular minichannel,
- the inner copper tube,
- the stainless-steel sheath of the cartridge heater, together with selected internal components required to capture axial heat conduction inside the heater,
- the outer borosilicate glass tube,
- the aluminum inlet and outlet nozzles for the fluid.

The main physical properties of the materials used in the numerical calculations are shown in Table 2.

**Table 2.** The main physical properties of the materials used in the numerical calculations.

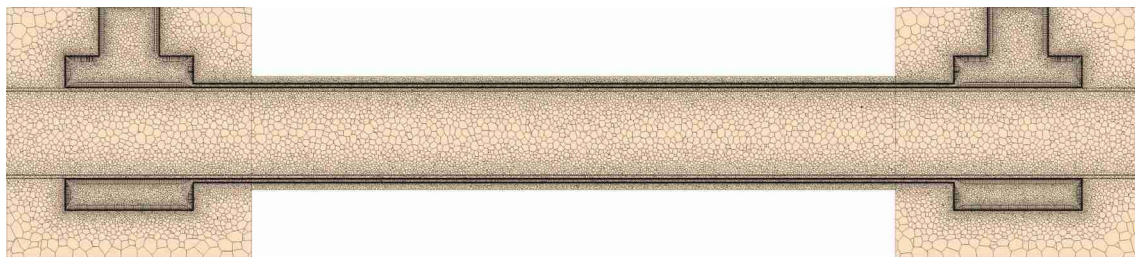
| Material        | Density<br>( $\text{kg}\cdot\text{m}^{-3}$ ) | Specific heat<br>( $\text{J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$ ) | Thermal conductivity<br>( $\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ ) | Dynamic viscosity<br>( $\text{kg}\cdot\text{m}^{-1}\cdot\text{s}^{-1}$ ) |
|-----------------|----------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------|
| HFE-649         | 1601.8                                       | 1103                                                                 | 0.0588                                                                     | 0.00064                                                                  |
| Aluminum        | 2702.0                                       | 903                                                                  | 237                                                                        | -                                                                        |
| Glass           | 2500.0                                       | 840                                                                  | 1.4                                                                        | -                                                                        |
| Stainless steel | 7850.0                                       | 486                                                                  | 52                                                                         | -                                                                        |
| Copper          | 8940.0                                       | 386                                                                  | 398                                                                        | -                                                                        |

**3.2. Mesh generation.** A polyhedral volume mesh was generated for the entire computational domain using the automatic mesher in Simcenter STAR-CCM+. Near the inner and outer walls of the annular gap, several layers of prismatic cells were employed to resolve the steep gradients of velocity and temperature, with particular attention paid to the thin annular region where most of the thermal resistance is concentrated. Number of elements in mesh, assumed in CFD calculation for both sets of heat flux, i.e. Set #1 (lower heat-input case - single phase flow regime) and Set #2 (higher heat-input case - the subcooled boiling region) are listed in Table 3.

**Table 3** Number of elements in mesh, Set #1 and Set #2

| Element | Minichannel | Fluid inlet nozzle | Fluid outlet nozzle | Glass tube | Copper tube | Cartridge heater |
|---------|-------------|--------------------|---------------------|------------|-------------|------------------|
| Set #1  |             |                    |                     |            |             |                  |
| cells   | 8 580 612   | 612 922            | 612 009             | 620 563    | 822 736     | 440 976          |
| faces   | 38 981 486  | 4 097 499          | 4 089 432           | 3 854 012  | 4 730 921   | 2 904 814        |
| verts   | 24 881 625  | 3 657 268          | 3 650 166           | 3 528 955  | 4 469 910   | 2 606 122        |
| Set #2  |             |                    |                     |            |             |                  |
| cells   | 10 271 773  | 789 900            | 800 115             | 665 726    | 1 831 398   | 388 124          |
| faces   | 49 260 646  | 5 320 999          | 5 386 664           | 4 132 227  | 11 388 757  | 2 515 416        |
| verts   | 33 498 411  | 4 738 350          | 4 797 819           | 3 787 352  | 10 431 574  | 2 269 572        |

A 3D digital model of the test module with the generated polyhedral mesh is shown in Fig. 4, which presents a longitudinal cross-section of the module.

**Fig. 4** The digital model of the test module with the generated polyhedral mesh.

Grid convergence was assessed using Roache's Grid Convergence Index (GCI) method, following the procedure described in [9] and applied in a similar manner to the authors' previous work [10]. The obtained GCI values indicated monotonic convergence and suggested that the solution is close to the asymptotic range.

**3.3. Governing equations and physical models.** The CFD simulations were performed in Simcenter STAR-CCM+ (Siemens Digital Industries Software) using the finite-volume method.

The fluid flow in a heated channel was analyzed for two selected heater power inputs described in Table 1. In both cases, the numerical simulations were carried out using the Finite Volume Method (FVM) [11]. The differences

between the two heater power inputs concerned the flow regime (laminar or turbulent) and the actual occurrence of a second phase. Regardless of the heater power inputs value, the governing description was based on the conservation equations of mass, momentum, and energy.

For the lower heater power input, the flow in the channel remained single-phase and laminar. A steady, three-dimensional flow of an incompressible Newtonian liquid was assumed. The flow was governed by the following equations [11]:

- continuity equation

$$\nabla \cdot (\rho \mathbf{v}) = 0 \quad (1)$$

- Navier–Stokes momentum equations

$$\nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v}) = \nabla \cdot (\mathbf{T} - p\mathbf{I}) + \mathbf{f}_b \quad (2)$$

- energy equation for the fluid

$$\nabla \cdot (\rho E \mathbf{v}) = \mathbf{f}_b \mathbf{v} + \nabla \cdot (\mathbf{v}(\mathbf{T} - p\mathbf{I})) - \nabla \cdot \mathbf{q} + S_E \quad (3)$$

Here,  $\mathbf{v}$  is the velocity vector,  $\mathbf{f}_b$  is the body-force term (gravity),  $p$  is the pressure,  $\rho$  is the density,  $\mu$  is the dynamic viscosity,  $\mathbf{T}$  is the viscous stress tensor,  $\mathbf{q}$  is the heat flux vector,  $E$  is the total energy per unit mass, and  $S_E$  is a volumetric energy source term.

Set #2 (the higher heater power input) corresponded to conditions reaching the subcooled boiling region, therefore an Eulerian multiphase formulation [11] was applied, in which the volume fractions of the liquid and vapor phases varied in space and satisfied the condition:

$$\alpha_l + \alpha_v = 1 \quad (4)$$

where  $\alpha_l$  is liquid volume fraction and  $\alpha_v$  is the vapor volume fraction.

The conservation equations of mass, momentum, and energy were solved for the mixture, with thermophysical properties evaluated as effective values dependent on the local phase volume fractions. To account for turbulence effects, the Reynolds-Averaged Navier–Stokes (RANS) [11] approach was adopted, in which each instantaneous variable  $\phi$  was decomposed into its mean ( $\bar{\phi}$ ) and fluctuating ( $\phi'$ ) components:

$$\phi = \bar{\phi} + \phi' \quad (5)$$

After Reynolds averaging, additional turbulent stress terms appeared in the momentum and energy equations. In the eddy-viscosity framework, the Reynolds stress tensor was expressed as

$$\mathbf{T}_{RANS} = \mu_t (\nabla \cdot \bar{\mathbf{v}} + \nabla \cdot \bar{\mathbf{v}}^T) - \frac{2}{3} k \mathbf{I} \quad (6)$$

where  $\mu_t$  is the turbulent eddy viscosity,  $k$  is the turbulent kinetic energy,  $\mathbf{I}$  is the identity tensor,  $\bar{\mathbf{v}}$  is the mean velocity vector. The turbulent viscosity  $\mu_t$  was determined using the  $k-\varepsilon$  turbulence model.

In the solid regions (copper tube, heater sheath and glass), only the stationary heat-conduction equation is solved. Conjugate heat transfer is enforced at all liquid–solid interfaces by imposing continuity of temperature and normal heat flux.

The dielectric working fluid HFE-649 is treated as incompressible. Its thermophysical properties ( $\rho$ ,  $\mu$ ,  $k$ ,  $c_p$ ) are taken from manufacturer data and/or standard material databases. Depending on the desired balance between accuracy and computational cost, properties may be specified either as constants evaluated at a representative

temperature or as temperature-dependent correlations. In the present work, constant thermophysical properties evaluated at a reference temperature  $T_{ref} = 298$  K were used.

Subsequently, the local heat transfer coefficient  $h$  in the annulus is evaluated in post-processing from the local wall heat flux and the wall-to-bulk fluid temperature difference (see Section 4.3).

**3.4. Boundary conditions.** At the inlet of the annular fluid domain, a velocity inlet boundary condition was imposed corresponding to the experimentally measured mass flow rate. The inlet temperature was set equal to the measured fluid temperature. At the outlet, a pressure outlet boundary condition was specified. In the heater domain, the electrical power supplied to the cartridge was represented as a volumetric energy source distributed within the heater components. The magnitude of the heat source was taken from the measured electrical power for each set.

Heat losses to the surroundings were represented by a convection boundary condition applied to the external glass surface with a constant heat transfer coefficient. The convective heat transfer coefficient was adopted from comparable cases reported in the literature [12,13] and set to  $h_{\infty} = 25$  W/(m<sup>2</sup>·K), and the ambient temperature taken from measurements. The measured ambient temperature was  $T_{\infty} = 293.15$  K.

All remaining solid–solid interfaces were treated as perfectly bonded, i.e. with no thermal contact resistance.

**3.5. Convergence criteria and solution procedure.** Steady-state solutions were obtained using a segregated solver with second-order spatial discretization for all transport equations. Under-relaxation factors were adjusted to ensure stable convergence for all operating points.

Convergence was assessed based on both residuals and monitored integral quantities. Iterations were continued until the scaled residuals of continuity and momentum decreased by at least three orders of magnitude, and the energy residual by at least four orders of magnitude. In addition, the following quantities were monitored:

- the total heat transferred to the fluid,
- the average wall temperature over the heated length,
- the axial wall-temperature profiles at thermocouple locations.

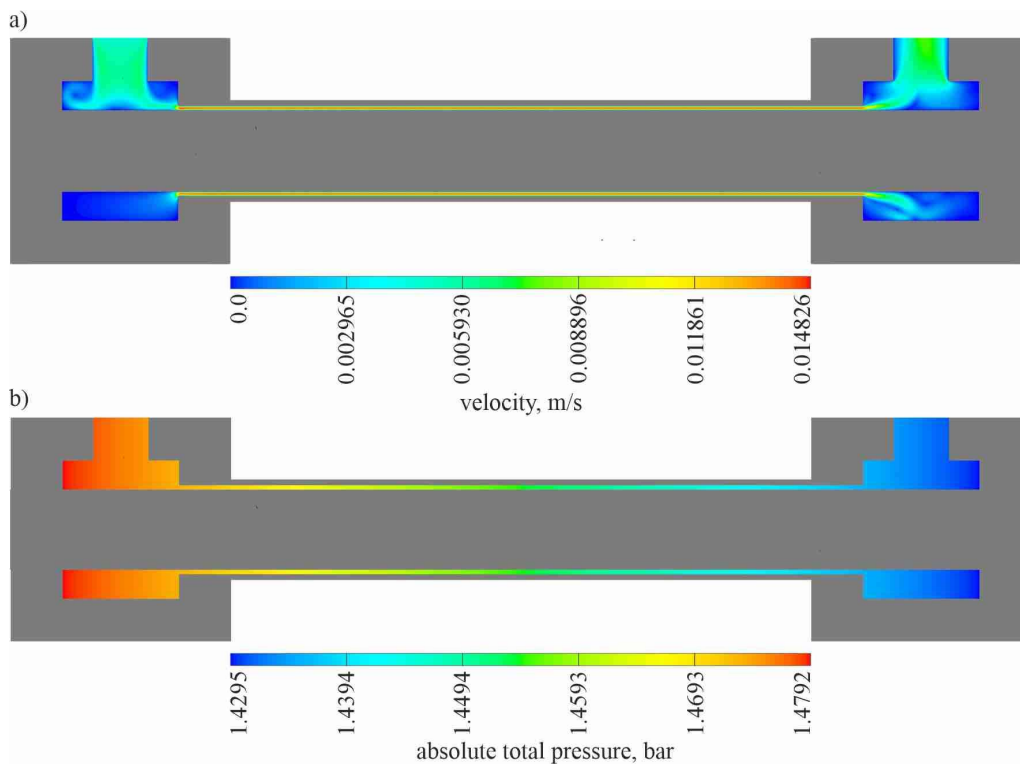
A solution was considered converged when these monitored quantities varied by less than a small tolerance over a sufficiently long iteration interval, e.g. a few hundred iterations.

## 4. RESULTS AND DISCUSSION

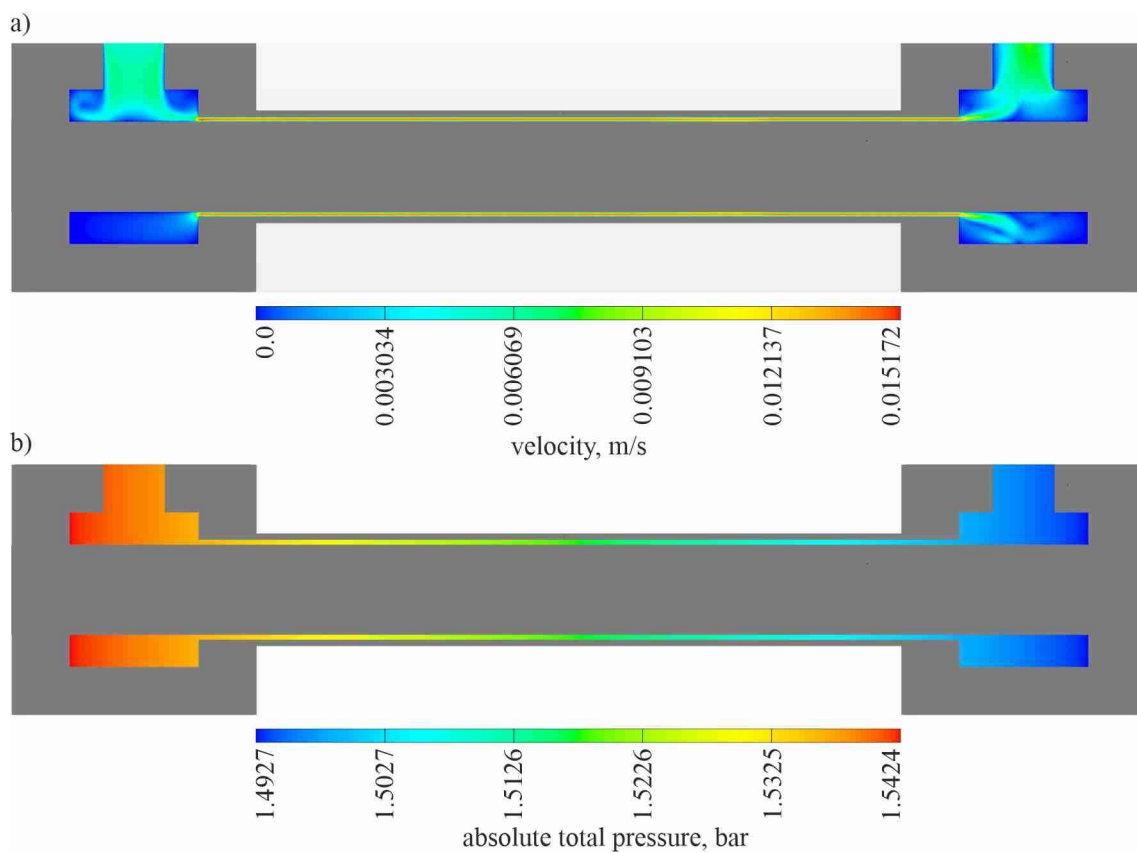
**4.1 Flow and pressure fields.** Figures 5 and 6 show the calculated distributions of velocity magnitude (Fig. 5a, 6a) and absolute pressure (Fig. 5b, 6b) in the fluid domain, respectively for Set #1 of heat flux value (Fig. 5) and Set #2 (Fig. 6).

When analyzing the results shown in Fig. 5 and Fig. 6 it was noted that:

- The velocity field exhibits the expected profile in the annular gap, with a nearly parabolic cross-sectional distribution and fully developed flow in the central part of the heated section. Entrance effects are visible near the inlet, where the velocity profile evolves from a top-hat shape to the fully developed form.
- The pressure decreases monotonically along the flow direction.

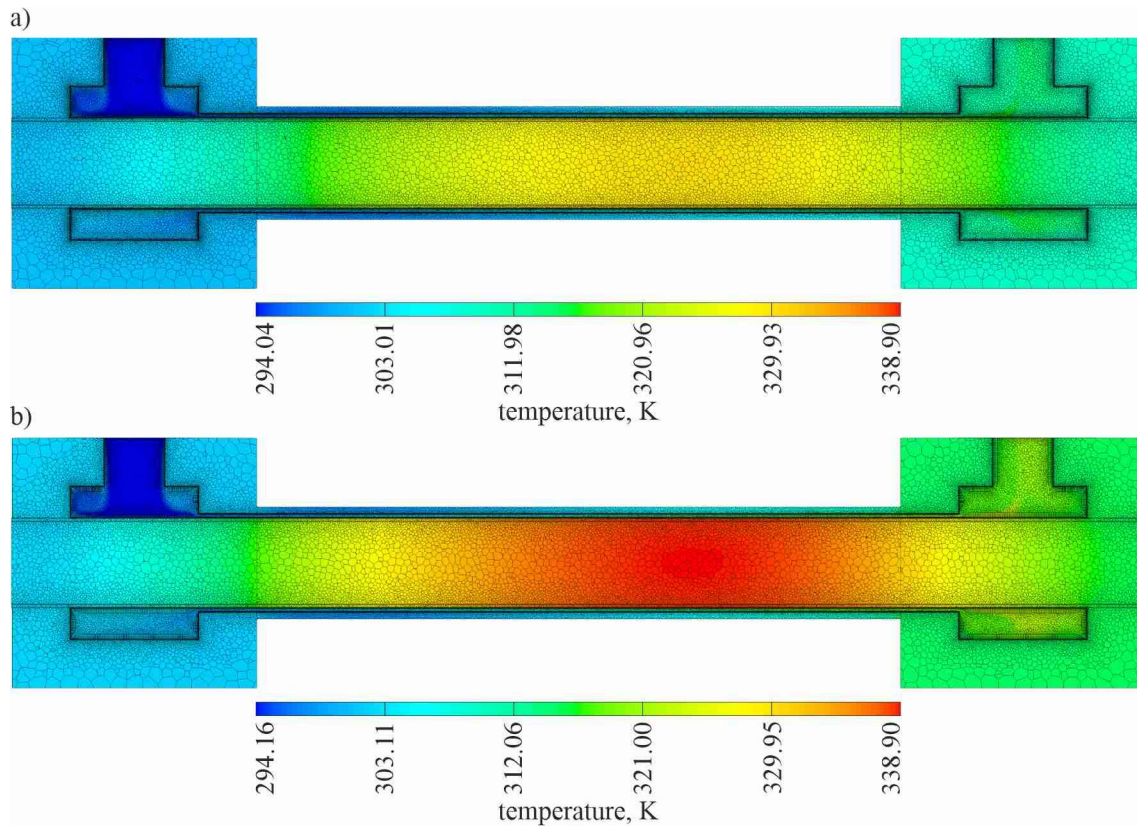


**Fig. 5** Distributions of velocity magnitude (a) and absolute pressure (b) in the fluid domain, Set #1.



**Fig. 6** Distributions of velocity magnitude (a) and absolute pressure (b) in the fluid domain, Set #2.

**4.2. Temperature distributions.** Figure 7 presents the temperature field in the entire test module and at the copper–fluid interface for Set #1 (Fig. 7a) and Set #2 (Fig. 7b), corresponding to two heater power inputs (two heat-flux levels). As expected for internal flow with uniform heating, the bulk fluid temperature increases monotonically along the axial direction; however, the heater temperature decreases in the outlet section due to end effects and enhanced heat removal in the outlet assembly/through external convection. The copper wall temperature generally rises downstream due to cumulative heat addition, and Set #2 exhibits higher wall and fluid temperatures in the mid-section as a result of the larger imposed heat input.

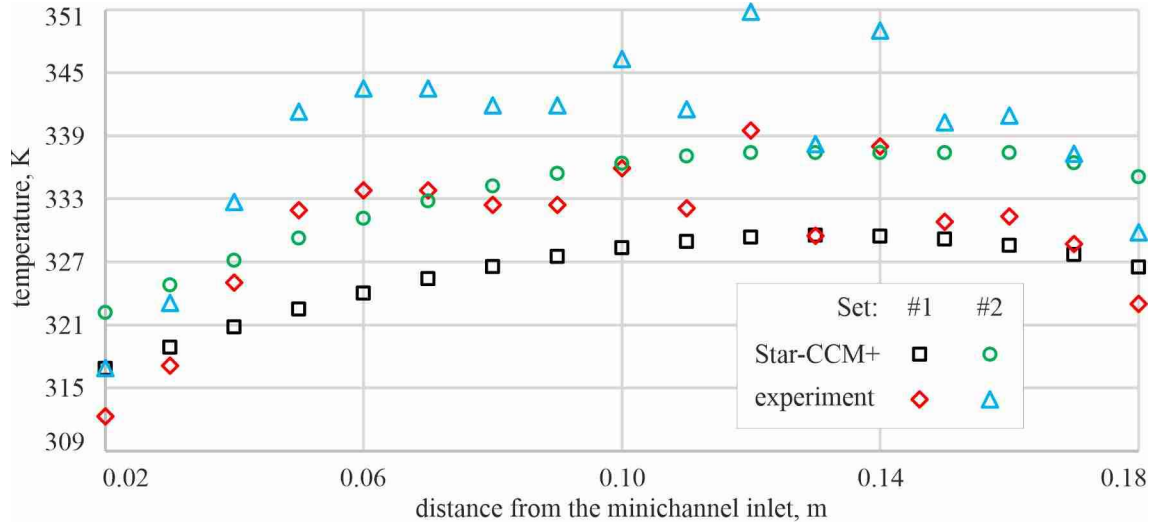


**Fig. 7** The temperature distribution in the entire test module as the results of CFD calculations in Simcenter STAR-CCM+, obtained for: a) Set #1, b) Set #2.

The axial wall-temperature profiles at the thermocouple locations for Set #1 and Set #2, obtained from the experiments and from CFD calculations in Simcenter STAR-CCM+, are illustrated in Fig. 8.

As shown in Fig. 8, at a fixed mass flow rate, increasing heater power (Set #2 versus Set #1) causes an increase the axial wall temperature along the entire minichannel length for both the experimental and numerical results. The experimental profiles exhibit a noticeably larger scatter than the CFD predictions. Figure 8 also indicates that the CFD model reproduces the overall shape of the measured wall-temperature profiles for both sets; however, except for the inlet and outlet regions, the numerical results are generally lower than the experimental values. The discrepancies are expected to be most pronounced in the developing regions near the inlet and outlet and can be attributed to uncertainty in thermocouple position and thermal contact, as well as to unavoidable heat losses to the surroundings that are difficult to represent exactly through boundary conditions. For the lower heat input (Set #1), the differences between the experimental and numerical results are larger than those observed for Set #2.

The relative differences between CFD-predicted and measured wall temperatures are discussed in detail in Section 4.4.



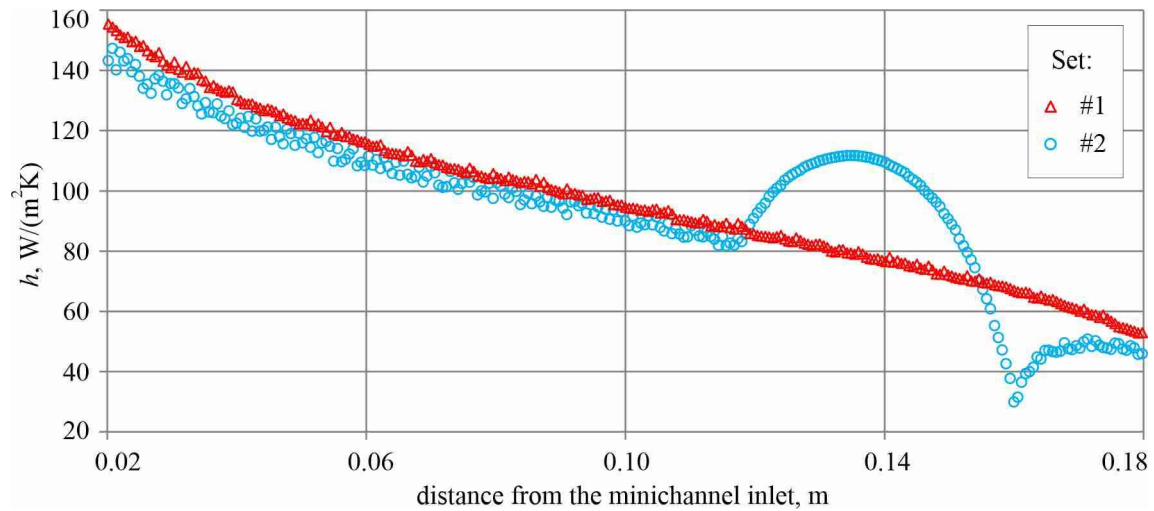
**Fig. 8** The axial profiles of wall temperature at the thermocouple locations, data for Set #1 and Set #2, obtained from experiment and from CFD calculations in Simcenter STAR-CCM+.

**4.3. Local heat transfer coefficients.** The local heat transfer coefficient was calculated in Simcenter STAR-CCM+ program as

$$h = \frac{q}{T_{wall} - T_{f,bulk}} \quad (7)$$

where  $q$  is the local wall heat flux,  $T_{wall}$  is the wall temperature,  $T_{f,bulk}$  is the local bulk temperature of the fluid. Both  $q$  and  $T_{wall}$  were extracted at the copper–fluid interface (fluid-side wall), whereas  $T_{f,bulk}$  denotes the bulk (mixed-mean) fluid temperature at the corresponding cross-section. Thus, the reported heat transfer coefficient represents the heat transfer predicted by the conjugate heat-transfer solution under the prescribed thermal input.

The axial distributions of  $h$  for Set #1 and Set #2 are shown in Fig. 9.



**Fig. 9** Axial distributions of  $h$  predicted by Simcenter STAR-CCM+ for Set #1 and Set #2.

For both heater power inputs, the heat transfer coefficient exhibits a pronounced entrance region with relatively high values of  $h$ . For the lower heat-input case (Set #1),  $h$  decreases monotonically along the channel and reaches its minimum near the outlet. Furthermore, for Set #2 the initial downstream decrease is followed (at approximately two-thirds of the heated length) by a distinct local increase in  $h$  and then by a sharp drop. Further downstream,  $h$  partially recovers and stabilizes at a nearly constant level close to the outlet, remaining slightly lower than the corresponding values for Set #1 in this region. The local increase and subsequent drop of  $h$ , observed for Set #2 are consistent with the onset of subcooled boiling (i.e., the transition from single-phase convection to boiling).

**4.4. Validation of the numerical simulations against experimental data.** In the final stage of the study, the CFD results were quantitatively compared with the experimental data. Validation was performed by comparing the CFD-predicted wall temperature at the thermocouple axial locations on the heater sheath with the measured thermocouple readings. It should be noted that the thermocouples do not measure the copper–fluid interface temperature directly. The comparison was carried out separately for the two heater power input sets (Set #1 and Set #2).

The agreement between CFD predictions and experiments was quantified using the relative difference defined by Eq. (8).

$$\delta_T = \left| \frac{T_{CFD,i} - T_{exp,i}}{T_{exp,i}} \right| \times 100\% \quad (8)$$

where  $\delta_T$  is the relative difference,  $T_{CFD,i}$  and  $T_{exp,i}$  denote the CFD-predicted and experimentally measured wall temperatures at the  $i$ -th thermocouple location.

The resulting values of  $\delta_T$  for two validation sets are summarized in Table 4.

**Table 4** Relative differences between the results obtained from numerical simulations in Simcenter STAR-CCM+ and the experimental data.

| Number of set                               | #1    | #2   |
|---------------------------------------------|-------|------|
| Min relative differences $\delta_T$ (%)     | 0.001 | 0.24 |
| Max relative differences $\delta_T$ (%)     | 3.58  | 3.82 |
| Average relative differences $\delta_T$ (%) | 1.66  | 2.05 |

The analysis of the results reported in Table 4 indicates that the relative differences between CFD-predicted and measured wall temperatures remain small for both heater power inputs. All relative differences values are lower for Set #1 in comparison to obtained for Set #2. This is consistent with the expectation that the smallest discrepancies occur in the single-phase convection regime (Set #1), whereas larger deviations may appear when phase-change effects become relevant, i.e., in the subcooled boiling region (Set #2). The maximum relative difference does not exceed 3.82% (for Set #2), while the average relative difference equals 1.66% for Set #1 and 2.05% for Set #2, indicating good consistency between the numerical predictions and the experimental data. The largest discrepancies tend to occur near the inlet/outlet regions, where the local flow and thermal fields are most sensitive to boundary-condition specification and where experimental uncertainty (e.g., thermocouple position) is higher.

Overall, the good agreement between CFD predictions and measured wall-temperature profiles indicates that the present three-dimensional conjugate-heat-transfer model can be reliably used as a predictive tool to analyze heat transfer during flow in annular minichannels.

## 5. CONCLUSIONS

The main conclusions from the numerical calculations are as follows:

(1) A three-dimensional CFD conjugate heat-transfer (CHT) model was developed in Simcenter STAR-CCM+ for HFE-649 flow in a vertical annular minichannel and assessed against experiments performed at a fixed mass flow rate of 8 kg/h for two heater power inputs (Set #1 and Set #2). The simulations were driven exclusively by measured boundary conditions (inlet fluid temperature, inlet/outlet pressures, mass flow rate, and heater power input); no measured wall temperatures were imposed.

(2) For both sets, the CFD predictions reproduce the overall shape and axial trend of the measured wall-temperature profiles at thermocouple locations. Outside the inlet/outlet developing regions, the predicted temperatures are generally lower than the experimental values, which is consistent with the combined influence of heat losses and measurement/model uncertainties (e.g., thermocouple positioning/contact and boundary-condition representation). The discrepancies are more pronounced for the lower heat input (Set #1) than for Set #2.

(3) Local heat transfer coefficients ( $h$ ) derived from the CFD fields exhibit a pronounced entrance region with relatively high values. For Set #1,  $h$  decreases monotonically along the channel and reaches its minimum near the outlet. For Set #2, the downstream decrease is followed by a distinct local increase and subsequent drop, after which  $h$  partially recovers and stabilizes toward the outlet; this behavior is consistent with the onset of subcooled boiling. The reported  $h$  distributions are mainly intended to assess axial trends and differences between the two sets, as the local values are sensitive to the definition of the bulk temperature and to small local values of  $T_{wall} - T_{f,bulk}$ .

(4) Overall, the combined experimental–numerical approach provides a consistent basis for interpreting temperature measurements in the tested annular minichannel configuration under controlled heating and offers a validated framework for further model development.

Further work should include completing the quantitative validation table, refining the treatment/estimation of heat losses in the thermal boundary conditions, and extending the dataset to additional operating points. The next steps will focus on extending the CFD approach to two-phase flow and boiling in the annular minichannel, as well as investigating alternative working fluids and modified geometries aimed at further intensification of heat transfer.

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## NOMENCLATURE

|            |                                      |                          |                  |                                    |                     |
|------------|--------------------------------------|--------------------------|------------------|------------------------------------|---------------------|
| $E$        | total energy per unit mass           | (J/kg)                   | $k$              | turbulent kinetic energy           | (J/kg)              |
| $f_b$      | body force per unit volume           | (N/m <sup>3</sup> )      | $p$              | pressure                           | (Pa)                |
| $h$        | heat transfer coefficient            | (W/(m <sup>2</sup> · K)) | $q$              | heat flux vector                   | (W/m <sup>2</sup> ) |
| $h_\infty$ | convective heat transfer coefficient | (W/(m <sup>2</sup> · K)) | Re               | Reynolds number                    | (–)                 |
| $I$        | identity tensor                      | (–)                      | $S_E$            | energy source term per unit volume | (W/m <sup>3</sup> ) |
|            |                                      |                          | $T$              | temperature                        | (K)                 |
|            |                                      |                          | $\boldsymbol{T}$ | viscous stress tensor              | (N/m <sup>2</sup> ) |

|              |                                 |                      |             |                                 |
|--------------|---------------------------------|----------------------|-------------|---------------------------------|
| $\mathbf{v}$ | velocity vector                 | (m/s)                | <i>CFD</i>  | CFD calculations                |
| $\nabla$     | nabla operator                  | (–)                  | <i>exp</i>  | experiment                      |
| $\alpha$     | volume fraction                 | (–)                  | <i>f</i>    | fluid                           |
| $\Delta T$   | temperature difference          | (K)                  | <i>l</i>    | liquid                          |
| $\mu$        | dynamic viscosity               | (kg/(m·s))           | <i>RANS</i> | Reynolds-Averaged Navier–Stokes |
| $\mu_t$      | turbulent eddy viscosity        | (kg/(m·s))           | <i>ref</i>  | reference                       |
| $\delta$     | average relative difference (%) |                      | <i>sat</i>  | saturation                      |
| $\rho$       | density                         | (kg/m <sup>3</sup> ) | <i>T</i>    | temperature                     |
|              |                                 |                      | <i>wall</i> | wall                            |
| Subscripts   |                                 |                      | <i>v</i>    | vapor                           |
| <i>abs</i>   | absolute                        |                      | $\infty$    | ambient temperature             |
| <i>bulk</i>  | bulk                            |                      |             |                                 |

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